University of Florence
School of Engineering - Department of Information Engineering

Thesis of the Master's Degree in Computer Engineering

Hybrid Quantum-Classical Architecture for Solving Large-Scale QUBO Problems: A Divide et Impera Approach

Candidate
Giacomo Orsucci

Supervisors
Prof. Benedetta Picano
Prof. Romano Fantacci

Co-supervisor
Dr. Claudio Cicconetti

Academic Year 2025/2026

Contents

Chapter 1

Introduction to Quantum Computing

In recent years, new and promising computing paradigms and architectures have emerged, complementing (and possibly competing with or outperforming in the near future) classical computing to solve highly complex problems. Among them, quantum computing stands out as one of the most promising and disruptive paradigms, expected to fundamentally revolutionize the way we process information.

This paradigm is profoundly different from classical computing, which uses a binary system of bits (0 or 1) to represent and process data. Quantum computing harnesses qubits, a completely different representation of information. Through quantum superposition, a qubit can exist simultaneously in a combination of states $|0\rangle$ and $|1\rangle$. Furthermore, quantum entanglement allows a deep correlation between qubits even over large distances, enabling the system to represent and process information in a fundamentally different way, with the potential to outperform classical computing in specific tasks [1, 2].

In this chapter, we will provide a brief overview of quantum computing, its evolution, and its promises. We will also compare different hardware paradigms, focusing on superconductors and neutral atoms, and discuss the characteristics and geometric constraints of neutral atom platforms.

1.1 Promises of quantum computing

The quantum computing paradigm currently promises to revolutionize information processing, offering the potential to solve problems that are intractable for classical computers. It is thus immediate to consider all the potential applications and fields that are currently facing the computational limits of NP-hard problems. While many heuristics, approximation algorithms, and metaheuristics have been developed to solve these problems, they are not always able to provide optimal solutions. Consequently, it is straightforward to recognize the potential of quantum computing in fields such as cryptography, optimization, machine learning, and drug discovery [3], where the ability to process large datasets and solve NP-hard problems on larger scales can lead to significant breakthroughs.

A comprehensive overview of the main promises of quantum computing (and neutral atoms in particular) can be found in [4], which discusses the primary characteristics and potential use cases of this technology. Additionally, a complete panorama of this paradigm, including its challenges, is detailed in [1]. Finally, to provide a summarized but complete introduction, we can conclude that quantum computing is a disruptive paradigm offering new opportunities for solving complex problems in various fields. However, it is still in its early stages of development, and many challenges remain to be addressed before it can reach its full potential.

These limitations and challenges must not be seen as a reason to avoid exploring this paradigm, but rather as a motivation to understand its boundaries and develop new algorithms capable of leveraging its unique capabilities. The importance of this matter is highlighted by the sector's market capitalization, estimated at approximately USD 10.13 billion in 2022 and projected to reach USD 125 billion by 2030 [1].

In the following sections, the main characteristics of neutral atom quantum computing are explored in more detail, together with their physical constraints, to provide a clear sense of why our exploration will face specific limitations in the next chapters of this thesis.

1.2 Hardware paradigms compared: Superconductors vs. Neutral Atoms

Currently, the quantum computing landscape is characterized by rapid technological advancement and intense competition. A diverse ecosystem of tech giants, research institutions, and startups is actively investigating various physical architectures. The shared objective is to develop a highly efficient and scalable computing paradigm capable of overcoming the fundamental challenges of the field, primarily quantum noise, limited coherence times, and hardware scalability [1].

To illustrate the diversity of this ecosystem, it is sufficient to look at the different technological pathways chosen by major industry players. Companies such as Google [5], Microsoft [6], and D-Wave [7] have heavily invested in superconducting qubits, whereas startups like Pasqal [8] and QuEra Computing [9] are pioneering platforms based on neutral atoms. Simultaneously, other organizations are exploring alternative technologies, including trapped ions, photonic circuits, silicon spin qubits, and innovative quantum error correction approaches, such as the hardware-efficient *cat qubits* developed by Alice & Bob [10], which aim to intrinsically mitigate decoherence and pave the way toward fault-tolerant architectures.

This section provides a comparative analysis of the two most prominent paradigms relevant to this work: superconducting circuits and neutral atoms.

1.2.1 Superconducting qubits

Without delving deeply into the physical details (which are beyond the scope of this thesis), superconducting qubits are based on Josephson junctions, superconducting circuits that exhibit quantum behavior at temperatures near absolute zero. These qubits are typically fabricated using advanced lithographic techniques and can be integrated into complex solid-state circuits. Superconducting qubits offer the advantage of very fast gate operations and relatively high fidelity, but they require extreme cryogenic cooling to maintain their quantum state [5].

A comprehensive overview of superconducting qubit technologies and the major players leading the field can be found in [3]. Currently, this type of qubit is the most mature and widely adopted in the industry; nevertheless, it faces significant challenges regarding scalability and coherence times. The strict requirement for cryogenic cooling and the engineering complexity of scaling up the number of qubits while maintaining high fidelity and mitigating cross-talk remain significant obstacles to achieving practical, large-scale quantum computing [3].

D-Wave is among the most well-known companies in this space, having developed commercial quantum annealers based on superconducting qubits designed specifically to solve optimization problems [7]. It is important to note that D-Wave's approach differs from the universal gate-based model pursued by Google and IBM, as it is tailored for specific optimization landscapes rather than general-purpose computing. However, for Quadratic Unconstrained Binary Optimization (QUBO) problems (which represent the core mathematical framework investigated in this thesis) quantum annealing remains one of the most effective approaches. While D-Wave's specific platform will not be extensively discussed in the remainder of this work, it stands as a commercially viable benchmark for solving large QUBO instances through purely quantum or hybrid approaches.

1.2.2 Neutral Atom platforms

Neutral atom platforms, on the other hand, utilize individual atoms trapped in optical lattices or optical tweezers to act as qubits. These atoms can be manipulated using finely tuned laser light, allowing their quantum states to be controlled with extreme precision. A primary advantage of neutral atom systems is their excellent isolation from the environment, which translates into long coherence times. Furthermore, they offer significant scalability potential, as large arrays of atoms can be trapped, dynamically positioned, and manipulated simultaneously. However, they also face significant engineering challenges, particularly regarding the complexity of controlling massive optical arrays and maintaining atomic stability within the traps over extended periods [3, 8].

These platforms are exceptionally promising for quantum simulation and combinatorial optimization tasks. By exciting the atoms to highly energetic states, they can naturally implement strong, long-range interactions that are difficult to achieve natively with other qubit technologies [8]. Due to this inherent physical capability, neutral atom architectures represent the primary focus of this thesis.

Among the main players in the field, Pasqal and QuEra Computing are notable for their pioneering work in developing these platforms. These companies have made significant strides in demonstrating the feasibility of large-scale neutral atom systems and their potential applicability in solving complex optimization problems [3, 8, 9].

Specifically, this work explores a hybrid quantum-classical architecture designed to tackle large-scale QUBO problems using these precise devices. Consequently, their capabilities and constraints are extensively investigated and discussed throughout the following chapters. As a foundation, the next section details the main characteristics and geometric constraints inherent to neutral atom platforms, which act as crucial parameters for the design of the hybrid architecture proposed in this thesis.

1.3 Characteristics of Neutral Atom platforms

This section focuses on the main characteristics and geometric constraints of neutral atom platforms, which are crucial for understanding the limitations and capabilities of these systems in the context of solving large-scale QUBO problems.

Neutral atom arrays A neutral atom array can be conceptualized as a hardware register where each individual atom acts as a qubit. These atoms are arranged in 1D, 2D (as in the context of this work), or 3D configurations [8, 11], as illustrated in Figure 1.1.

Unlike superconducting systems, where qubits are statically fabricated on a physical chip, neutral atoms can be dynamically arranged and manipulated using optical tweezers. This inherent flexibility allows for the creation

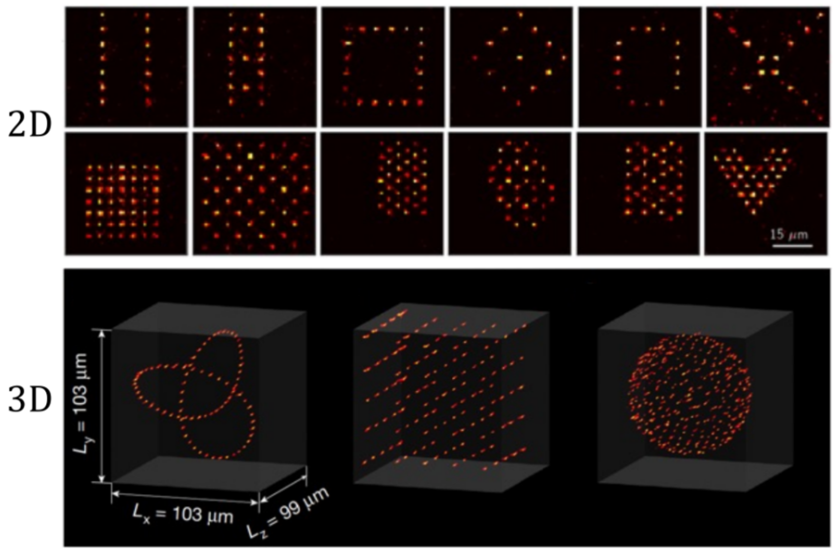

Figure 1.1: 2D and 3D neutral atom arrays. Image extracted from [8].

of custom geometries and the implementation of specific interaction topologies between qubits, which is highly advantageous for solving optimization problems that demand tailored connectivity.

However, because the register is not permanently fixed, it must be reconstructed for each computation cycle. The overall execution process consists of three main phases:

- Register preparation
- Quantum processing
- Register readout

The primary focus of this thesis, however, precedes the physical execution phase. It centers on the algorithmic challenge of efficiently embedding a QUBO problem onto this spatially constrained register to find optimal or near-optimal solutions. While the remainder of this work focuses mainly on the algorithmic approach, leveraging libraries and remote simulation infrastructures rather than delving into low-level hardware control, understanding the physical constraints of the QPU is essential. Therefore, following the explanations provided by [8], we briefly outline the physical mechanisms behind register loading and readout.

Register loading Configuring the atomic register requires a complex optical setup, relying on arrays of optical tweezers alongside the components illustrated in Figure 1.2.

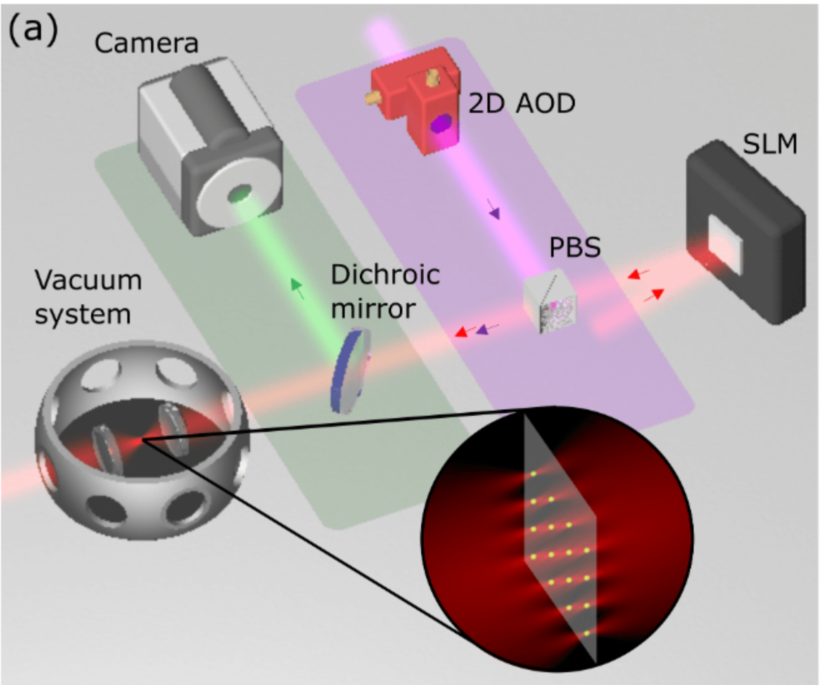

Figure 1.2: Main hardware components of a neutral atom QPU. The trapping laser light (red) is shaped by a Spatial Light Modulator (SLM). Moving tweezers (purple), used to rearrange atoms, are controlled by a 2D Acousto-Optic Deflector (AOD) and superimposed using a Polarizing Beam-Splitter (PBS). The green fluorescence emitted by the atoms is separated by a dichroic mirror and captured by a camera during the readout phase. Image adapted from [8].

In essence, the process begins inside a vacuum chamber, where a laser system cools and confines a cloud of atoms. Subsequently, high-precision lenses focus light beams into extremely tight spots, creating optical tweezers. Each optical trap is designed to isolate and hold a single atom. By leveraging holographic techniques through a Spatial Light Modulator (SLM), it is possible to project these tweezers into predefined positions, generating the desired geometric array. This mechanism is the cornerstone of the technology's scalability: theoretically, scaling the system primarily requires increasing laser power and improving optical resolution to project more traps.

However, loading atoms from the cooled cloud into the traps is a stochastic process. Due to light-assisted collisions, each optical tweezer has roughly a 50% probability of capturing exactly one atom, leaving the remaining traps empty. To construct a defect-free computational register, the system must perform an active rearrangement. This is achieved using dynamic tweezers governed by Acousto-Optic Deflectors (AODs), which sort and move atoms from filled traps into empty ones. This sorting process is repeated until the target array is completely filled, resulting in a perfect register ready for the quantum processing phase [8].

Quantum processing and register readout Once the atomic array is perfectly assembled, the quantum processing phase can begin. It is worth noting a significant timescale disparity in this hardware: while the actual quantum computation (performed via precisely timed laser pulses) is extremely fast, typically lasting less than 100 µs, the entire computational cycle is dominated by the physical loading and readout phases, requiring approximately 200 ms in total [8].

After the algorithm is executed, the final state of the register must be measured to extract the classical solution. This readout phase is performed through fluorescence imaging. The physical system is tuned so that atoms remaining in the ground state (representing the logical qubit state $|0\rangle$) emit photons and appear as bright spots on the image. Conversely, atoms excited to the Rydberg state (representing the logical qubit state $|1\rangle$) do not fluoresce and remain dark.

These optical signals are captured by highly sensitive Electron-Multiplying CCD (EMCCD) cameras. This detection method is extremely reliable, reaching readout efficiencies above 98.6%, a performance highly competitive with the readout fidelities typical of superconducting circuits [8]. Finally, due to the inherently probabilistic nature of quantum mechanics, a single execution yields only one possible classical bit-string. Therefore, the complete sequence of preparation, processing, and readout must be repeated hundreds or thousands of times (taking multiple *shots*) to reconstruct the statistical probability distribution of the outcomes and successfully identify the optimal solution.

Quantum analog mode execution Neutral atom platforms can operate in both digital and analog modes. This work concentrates exclusively on the latter; please refer to [8] for more information on the former. The dynamics of the analog mode are well-described by the Ising model, governed by the following Hamiltonian:

$$\hat{\mathcal{H}}(t) = \hbar \sum_{i=1}^{N} \left(\frac{\Omega(t)}{2} \hat{\sigma}_i^x - \delta(t) \hat{n}_i \right) + \sum_{i<j} \frac{C_6}{|r_i - r_j|^6} \hat{n}_i \hat{n}_j \tag{1.1}$$

This equation can be conceptually divided into two parts. The first summation dictates the evolution of single atoms driven by the external laser field, while the second summation governs the interactions between pairs of atoms. Specifically, the parameters and operators are defined as follows:

- N is the total number of atoms (qubits) in the register;

- $\Omega(t)$ is the time-dependent Rabi frequency, representing the amplitude of the laser exciting the atoms;
- $\delta(t)$ is the detuning parameter of the laser frequency;
- $\hat{\sigma}_i^x$ is the Pauli-X operator acting on atom i, which drives the transition between the ground state $|0\rangle$ and the Rydberg state $|1\rangle$;
- $\hat{n}_i$ is the occupation number operator (defined as $|1\rangle_i\langle 1|_i$) which evaluates to 1 if atom i is in the Rydberg state, and 0 otherwise;
- r_i is the spatial position of atom i;
- C_6 is the Van der Waals interaction coefficient, which is a constant dependent on the specific atomic species and platform used [12].

Crucially, the third term in this Hamiltonian represents the interaction between individual spins. It accounts for the energy penalty incurred when two spatially close qubits occupy the Rydberg state simultaneously. When excited, qubits transition to the Rydberg state; however, if they are physically in close proximity, they experience the *Rydberg blockade effect.* In this regime, maintaining the Rydberg state by the first qubit heavily suppresses the excitation of the second. Qubits in Rydberg states interact with each other proportionally to the third term above, following the Van der Waals potential for dipole-dipole interactions:

$$V_{ij} = \frac{C_6}{r_{ij}^6} \tag{1.2}$$

where:

- V_{ij} is the interaction energy between atoms i and j;

- C_6 is the Van der Waals coefficient, which depends on the atom type and scales massively with the excitation level;
- r_{ij} is the spatial distance between the two atoms [8].

Chapter 2

The Embedding Problem: Theoretical Framework

This chapter introduces the theoretical framework behind the embedding problem for Neutral Atom Quantum Architectures. The aim is to explain the inherent complexity of finding efficient embeddings. We will start by introducing basic concepts of computational complexity and graph theory, and then proceed to define Unit Disk Graphs (UDG) and the Maximum Independent Set (MIS) problem. Finally, these concepts will be connected to the current State of the Art (SOTA) and the Quadratic Unconstrained Binary Optimization (QUBO) formulation.

2.1 Combinatorial Optimization and Computational and Geometrical Complexity

Combinatorial optimization is a prominent branch of applied mathematics and computer science that focuses on finding an optimal object from a finite, discrete set of possible solutions. Specifically, the objective is to identify a solution that minimizes (or maximizes) a well-defined cost function, often subject to a set of strict mathematical constraints [13].

These problems commonly involve discrete mathematical structures such as graphs, permutations, or boolean variables and are ubiquitous in fields ranging from logistics and network routing to machine learning and quantum physics [14]. Although the solution space for these problems can be strictly finite, the number of possible configurations still can be overwhelmingly large.

Moreover, combinatorial optimization frequently encounters NP-Hard problems, where finding the exact global optimum becomes computationally intractable. Consequently, the field heavily leverages heuristics and metaheuristics to find high-quality approximate solutions within a reasonable timeframe. Within this computationally demanding landscape, Quantum Computing emerges as a disruptive paradigm; by exploiting quantum mechanical phenomena, it promises to tackle significantly larger NP-Hard instances and achieve higher-quality approximate solutions more efficiently than classical approaches [3].

Building upon this general computational complexity, a secondary, equally formidable challenge arises when utilizing neutral atom platforms: the geometrical complexity. As introduced in Chapter 1, executing an algorithm requires mapping (or *embedding*) the logical variables of the optimization problem onto the physical atoms arranged in a 2D or 3D register [8]. In our specific case, focusing on a 2D plane, the hardware imposes strict geometric constraints and specific interaction boundaries governed by the Ising model and the Rydberg blockade. Finding an efficient spatial mapping that preserves the problem's theoretical connectivity while respecting these strict physical limitations is, remarkably, an NP-Hard problem in itself [11].

2.2 Unit Disk Graphs (UDG) and the Maximum Independent Set (MIS)

A graph can be formally denoted as:

$$G = (V, E) \tag{2.1}$$

where V is the set of vertices (or nodes) and E is the set of edges connecting them. Graphs are ubiquitous mathematical structures used to model pairwise relations between objects, and their properties are supported by a vast and robust theoretical foundation [15].

Consequently, while an exhaustive review of graph theory is beyond the scope of this work, these structures are of fundamental interest. Indeed, a significant portion of current literature focuses on graph-based combinatorial problems, primarily because graphs serve as the standard mathematical interface to encode and map optimization tasks onto quantum architectures. Prominent examples of such NP-Hard problems include the Maximum Cut (Max-Cut), the Traveling Salesperson Problem (TSP), and the Maximum Independent Set (MIS), which have extensively served as standard benchmarks for evaluating the performance of quantum algorithms [16].

To specialize our focus from general graphs, we introduce the concept of Unit Disk Graphs (UDG). As formalized by Clark et al. [17], UDGs can be defined through three mathematically equivalent geometric models:

- **Distance Model:** Given a set of n points in the Euclidean plane, an n-vertex graph is formed where each vertex corresponds to a point ($|V| = n$). An edge exists between two vertices if and only if the Euclidean distance $d(u, v)$ between the two corresponding points u and

v is at most a specified threshold radius r. Mathematically, the edge set is defined as:

$$E = \{\{u, v\} \mid u, v \in V,\ d(u, v) \leq r\} \tag{2.2}$$

- **Intersection Model:** Consider a set of n equal-sized circles in the plane. The UDG is the intersection graph of these circles: each vertex corresponds to a circle, and an edge appears between two vertices if their corresponding circles overlap (tangent circles are also assumed to intersect).
- **Containment Model:** Alternatively, considering again n equal-sized circles, an edge exists between two vertices if the circle corresponding to one vertex geometrically contains the center of the other.

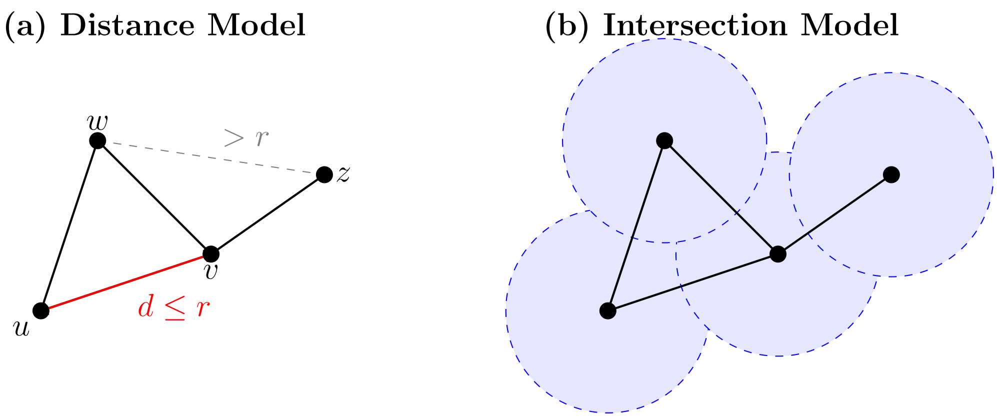

Figure 2.1: Equivalent geometric representations of a Unit Disk Graph (UDG). **(a)** The Distance Model connects nodes u and v if their Euclidean distance is $d \leq r$. **(b)** The Intersection Model places a disk of radius $r/2$ centered on each node; edges are formed exclusively where disks overlap.

Given any of these three definitions, the other two can be derived in linear time, demonstrating that they describe the exact same class of graphs. Historically, UDGs have proven to be an excellent mathematical framework for modeling broadcast and cellular networks, effectively resolving real-world problems such as frequency assignment and emergency signal routing [17].

In the context of this thesis, the distance model is of paramount importance: replacing the generic threshold distance r with the Rydberg radius (R_b) perfectly captures the physical interaction connectivity of neutral atoms in a quantum register.

As previously mentioned, the Maximum Independent Set (MIS) is a classic NP-Hard combinatorial optimization problem. While canonically formulated on general graphs, it holds massive relevance for our architecture because the problem famously remains NP-Hard even when restricted to Unit Disk Graphs [17].

Formally, given a graph $G = (V, E)$, an *Independent Set* (IS) is defined as a subset of vertices $I \subseteq V$ such that no two vertices within I are adjacent. Mathematically, this constraint translates to:

$$\forall u, v \in I, \quad \{u, v\} \notin E \tag{2.3}$$

The MIS problem entails finding an independent set that contains the largest possible number of vertices; in other words, the objective is to maximize the cardinality $|I|$.

Given the aforementioned definitions, a straightforward and perfect isomorphism emerges between the abstract MIS problem on a UDG and the physical dynamics of neutral atom arrays. Specifically, the geometric threshold radius r of the UDG corresponds directly to the Rydberg blockade radius (R_b).

Consequently, the core mathematical constraint of the MIS problem, which strictly forbids selecting two adjacent vertices, is natively and physically enforced by the Rydberg blockade phenomenon, which prevents any two atoms within a distance R_b from being simultaneously excited to the Rydberg state. Because of this perfect theoretical alignment, applying a global laser pulse designed to maximize the number of excited atoms in the 2D register forces the quantum many-body system to naturally evolve toward a ground state that inherently encodes the exact solution to the Maximum Independent Set of the corresponding geometric graph [8].

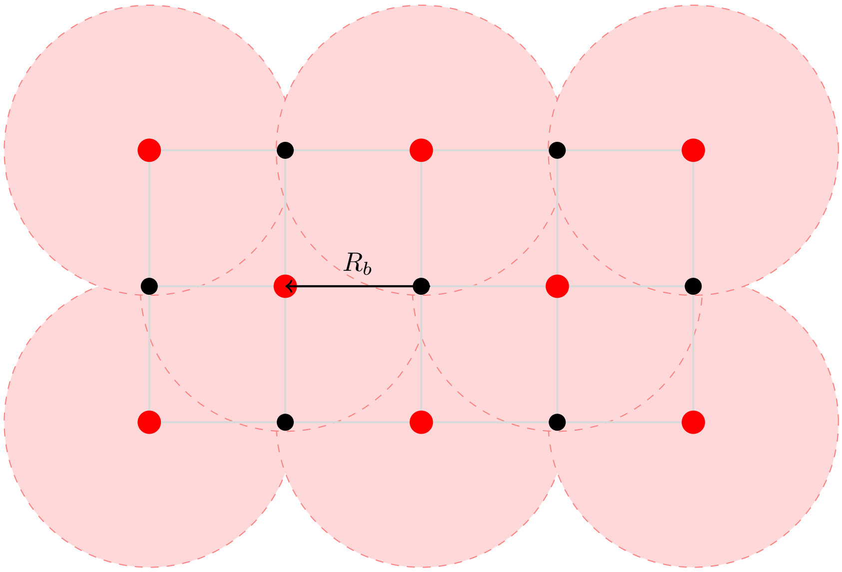

Figure 2.2: Physical realization of the Maximum Independent Set (MIS) via the Rydberg blockade. Red dots represent atoms excited to the Rydberg state (**MIS selected nodes**), while black dots are atoms in the ground state (**unselected nodes**). The shaded red regions represent the Rydberg blockade volume (radius R_b). The physical mechanism preventing two Rydberg excitations from overlapping perfectly enforces the mathematical constraint of the MIS problem.

2.3 Literature and State of the Art

Defining a definitive State of the Art (SOTA) for the embedding problem on neutral atom architectures is inherently challenging, as the field remains at the absolute frontier of quantum computing research. Currently, there is no universally established protocol or singular "gold standard" architecture. Instead, the literature presents a diverse landscape of experimental approaches and heuristic strategies aimed at overcoming the strict geometric constraints of these platforms.

To illustrate this dynamic ecosystem, recent literature explores a wide array of embedding techniques. For instance, Vercellino et al. [11] investigate the use of Neural Networks to dynamically map problem graphs onto physical registers. Other approaches rely on force-directed heuristics; notably, Tarquini et al. [18] successfully experimented with the Fruchterman-Reingold (FR) algorithm, enhanced by sophisticated pre- and post-processing techniques, to embed Quadratic Unconstrained Binary Optimization (QUBO) problems.

Alongside embedding algorithms, significant research is devoted to translating highly complex real-world use cases into physical layouts. Examples include the formulation of emergency hub placement routing [19] and satellite mission planning [12], demonstrating the vast applicability of QUBO formulations on neutral atoms.

Crucially for this thesis, several recent studies have started exploring problem partitioning as a workaround for hardware scale limitations. Works by Nembrini et al. [20, 21] and Das et al. [22] propose dividing the original combinatorial problem into sub-instances resolved using localized arrays. While their specific methodologies differ from our scope, these partitioning strategies served as the foundational inspiration for the hybrid *divide et impera* (divide and conquer) architecture proposed in the subsequent chapters of this work.

This diversity of approaches underscores the immaturity and the highly exploratory nature of the field. While the lack of rigid standards offers significant opportunities for novel theoretical contributions, it also introduces substantial experimental risks and engineering bottlenecks.

Drawing from this extensive body of literature, we elected to restrict our focus specifically to the QUBO mathematical framework, which will be formally defined in the next section. More importantly, driven by the operational realities and constraints of the currently available experimental platforms, our investigation will target QUBO instances that naturally exhibit a UDG-like connectivity. This specific focus bypasses the need for massive auxiliary embeddings, setting the optimal stage for our distributed algorithmic approach.

Bibliography

[1] Q. A. Memon, M. A. Ahmad, and M. Pecht, "Quantum computing: Navigating the future of computation, challenges, and technological breakthroughs," *Quantum Reports*, vol. 6, no. 4, pp. 627–663, 2024.

[2] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*. Cambridge University Press, 10th anniversary ed., 2010.

[3] Q. A. Memon, M. Al Ahmad, and M. Pecht, "Quantum computing: Navigating the future of computation, challenges, and technological breakthroughs," *Quantum Reports*, vol. 6, pp. 627–663, 2024.

[4] M. Grotti, S. Marzella, G. Bettonte, D. Ottaviani, and E. Ercolessi, "Practical use cases of neutral atoms quantum computers," 2026.

[5] Google, "What is quantum computing?," whitepaper, Google, 2024. Guide for policymakers.

[6] Microsoft, "Microsoft's majorana 1 chip carves new path for quantum computing," 2024.

[7] C. C. McGeoch and P. Farré, "The d-wave advantage system: An overview," Technical Report 14-1049A-A, D-Wave Systems Inc., 2020.

[8] L. Henriet, L. Beguin, A. Signoles, T. Lahaye, A. Browaeys, G.-O. Reymond, and C. Jurczak, "Quantum computing with neutral atoms," *Quantum*, vol. 4, p. 327, Sept. 2020.

[9] S. Ebadi, A. Keesling, M. Cain, T. T. Wang, H. Levine, D. Bluvstein, G. Semeghini, A. Omran, J.-G. Liu, R. Samajdar, X.-Z. Luo, B. Nash, X. Gao, B. Barak, E. Farhi, S. Sachdev, N. Gemelke, L. Zhou, S. Choi, H. Pichler, S.-T. Wang, M. Greiner, V. Vuletic, and M. D. Lukin, "Quantum optimization of maximum independent set using rydberg atom arrays," *Science*, vol. 376, no. 6598, pp. 1209–1215, 2022.

[10] N. Coppola, R. Lescanne, L. Prost, and M. Žeško, "Think inside the box: Quantum computing with cat qubits," tech. rep., Alice & Bob, December 2024. Curated by Brian Siegelwax.

[11] C. Vercellino, P. Viviani, G. Vitali, A. Scionti, A. Scarabosio, O. Terzo, E. Giusto, and B. Montrucchio, "Neural-powered unit disk graph embedding: qubits connectivity for some qubo problems," in *2022 IEEE International Conference on Quantum Computing and Engineering (QCE)*, p. 186–196, IEEE, 2022.

[12] M. Nowak, B. Marchand, Y. Naghmouchi, S. Rainjonneau, W. Coelho, L. Vignoli, and L.-P. Henry, "Satellite mission planning with rydberg atoms," 2026.

[13] B. Korte and J. Vygen, *Combinatorial Optimization: Theory and Algorithms*. Springer, 5th ed., 2012.

[14] C. H. Papadimitriou and K. Steiglitz, *Combinatorial Optimization: Algorithms and Complexity*. Dover Publications, 1998.

[15] C. Berge, *The theory of graphs.* Courier Corporation, 2001.

[16] A. Lucas, "Ising formulations of many np problems," *Frontiers in Physics*, vol. 2, p. 5, 2014.

[17] B. N. Clark, C. J. Colbourn, and D. S. Johnson, "Unit disk graphs," *Discrete Mathematics*, vol. 86, no. 1, pp. 165–177, 1990.

[18] S. Tarquini, M. Vandelli, F. Ferrari, D. Dragoni, and F. Tudisco, "Drone delivery packing problem on a neutral-atom quantum computer," 2026.

[19] S. Tarquini, M. Vandelli, F. Ferrari, D. Dragoni, and F. Tudisco, "Emergency hub placement with a neutral-atom quantum computer," 2026.

[20] R. Nembrini, M. Ferrari Dacrema, and P. Cremonesi, "An application of reinforcement learning for minor embedding in quantum annealing," in *Proceedings of the 2nd International Workshop on AI for Quantum and Quantum for AI (AIQxQIA 2024) co-located with the 23rd International Conference of the Italian Association for Artificial Intelligence (AIxIA 2024)*, vol. 3913 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2024.

[21] R. Nembrini, M. Ferrari Dacrema, and P. Cremonesi, "Minor embedding for quantum annealing with reinforcement learning," *Quantum Machine Intelligence*, vol. 8, Feb. 2026.

[22] S. Das, S. K. Roy, R. Rana, and M. G. Chandra, "Grid-partitioned mwis solving with neutral atom quantum computing for qubo problems," *arXiv preprint arXiv:2510.18540*, 2025.